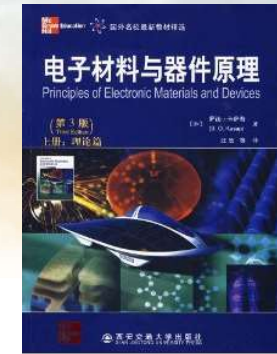
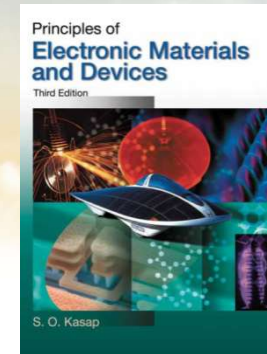




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch4-3 Modern Theory of Solids

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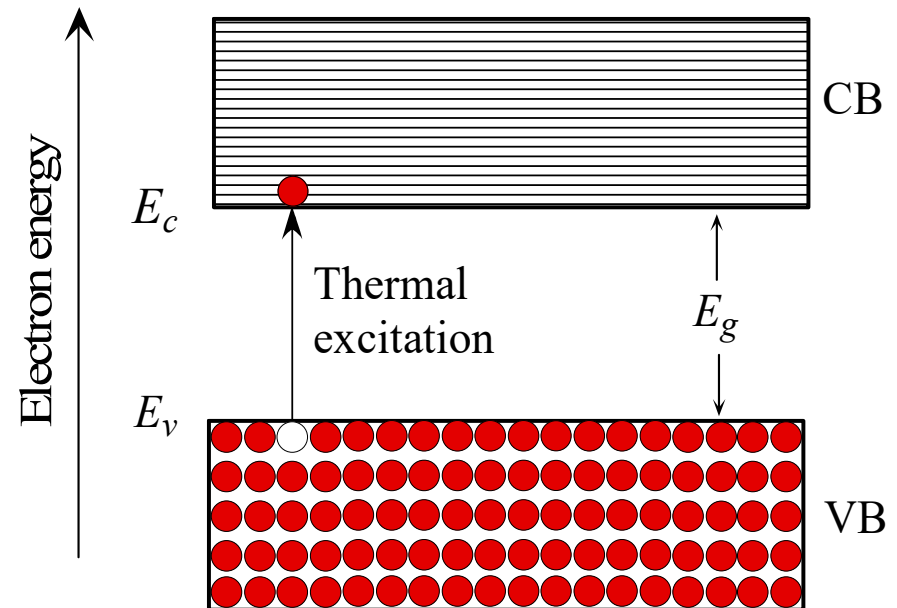


Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

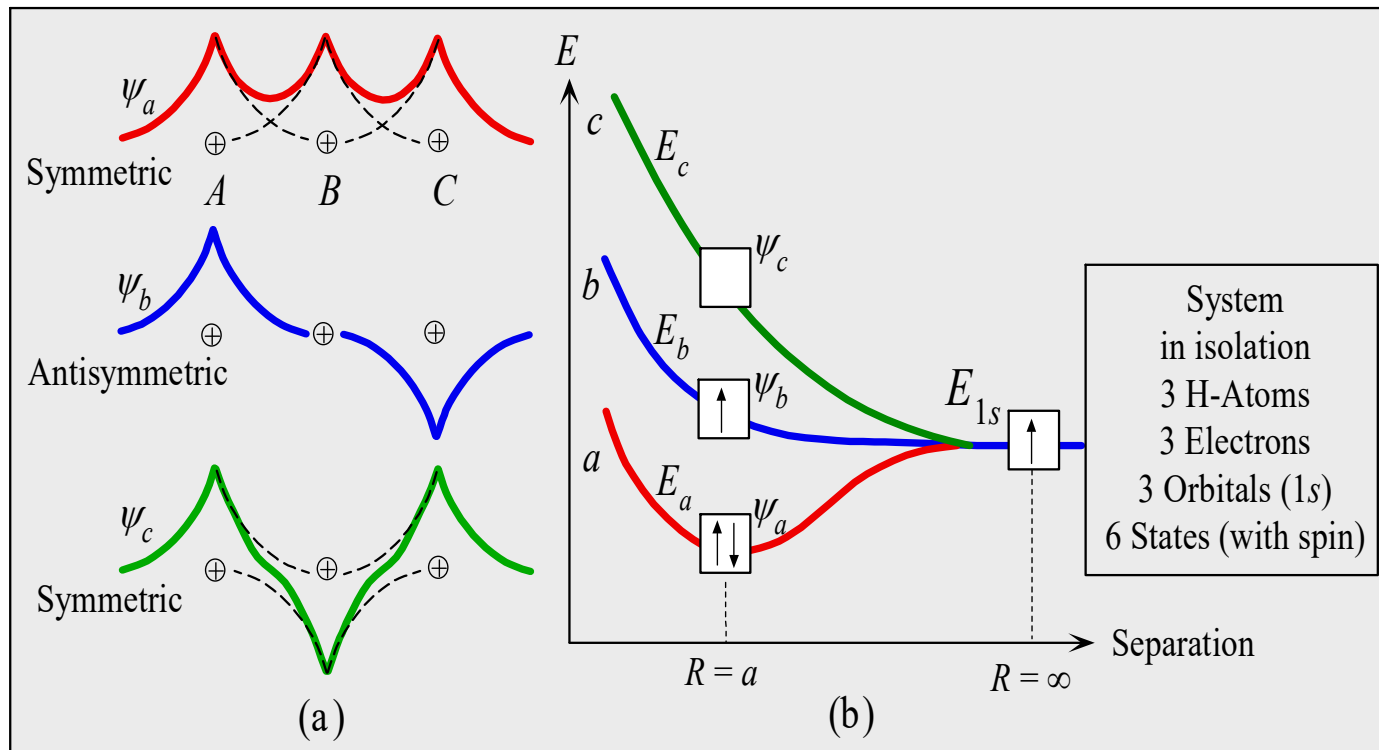
Introduction of Density of State (DOS)

Questions:

- The CB and VB can theoretically accommodate how many electrons?
A: Related to density of states
- Is doping concentration the larger the better?
A: Related to density of states
- How many electrons are actually in the bands?
A: Related to density of states



In the Case of H₃

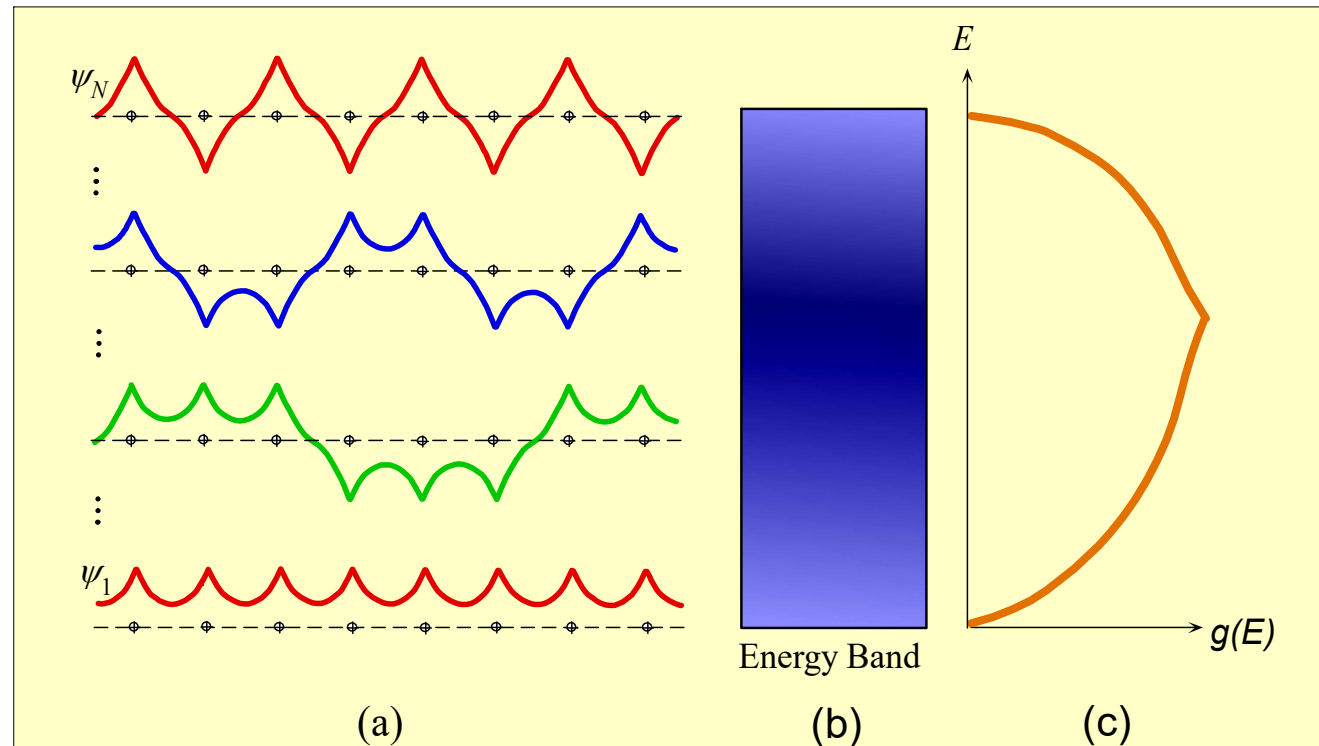


- When three H atoms are brought together, the potential energy for A, B, C atom is different, depending on their position.
- Resulting in three different wave functions
- Electrons sit the orbital with lower energy first

$$\begin{aligned}\psi_a &= \psi_{1s}(A) + \psi_{1s}(B) + \psi_{1s}(C) \\ \psi_b &= \psi_{1s}(A) - \psi_{1s}(C) \\ \psi_c &= \psi_{1s}(A) - \psi_{1s}(B) + \psi_{1s}(C)\end{aligned}$$



Density of State



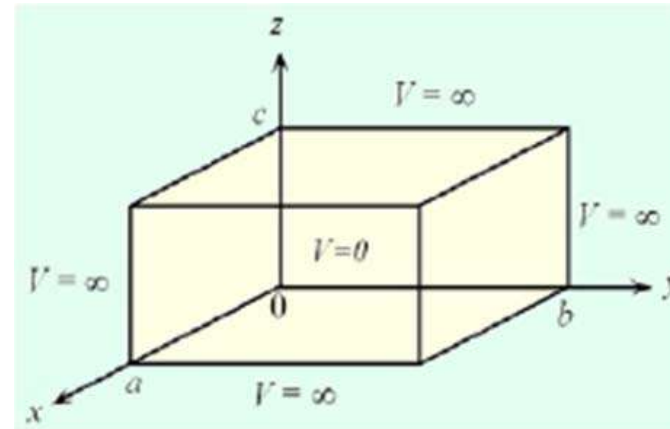
- Lower energy: all wave functions are in phase
- Higher energy: all wave functions are out of phase
- Energy state $g(E)$ is the number of states in the energy interval E to $E+dE$ per unit volume
- The **number of states per unit volume up to energy E'** is $S_v(E') = \int_0^{E'} g(E) dE$



Potential Box: Three Quantum Numbers

To examine the properties of a particle confined to a region of space, we take a 3D space with a volume marked by a, b, c along the x, y, z axes. The PE is zero inside the space and infinite on the outside.

What is the behavior of the electron in this box?



3-D version of the Schrödinger equation

$$\frac{d^2\varphi}{dx^2} + \frac{d^2\varphi}{dy^2} + \frac{d^2\varphi}{dz^2} + \frac{2m}{\hbar^2}(E - V)\varphi = 0$$

- We can solve this by separating the variables via $\varphi(x, y, z) = \varphi(x)\varphi(y)\varphi(z)$
and total simple wave-function is simply the product $\varphi(x, y, z) = A \sin(k_x x) \sin(k_y y) \sin(k_z z)$
- Applying the boundary conditions at $x=a, y=b, z=c$ to determine the constants k_x, k_y, k_z gives

$$k_x = \frac{n_1\pi}{a} \quad k_y = \frac{n_2\pi}{b} \quad k_z = \frac{n_3\pi}{c}$$



Potential Box: Three Quantum Numbers

- Eigenfunction of the electron, denoted by the quantum number n_1, n_2, n_3 , are now given by

$$\varphi_{n_1 n_2 n_3}(x, y, z) = A \sin\left(\frac{n_1 \pi x}{a}\right) \sin\left(\frac{n_2 \pi y}{b}\right) \sin\left(\frac{n_3 \pi z}{c}\right)$$



These wavefunctions consist of the products of infinite 1-D PE well-type wave-functions, one for each dimensions, and each has its own quantum number n



Each possible eigenfunction can also be labeled a state for the electron

- We can find the energy of the electron by substituting the wave-function into (1)

$$E = E(k_x, k_y, k_z) = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2)$$

and using (2)

$$E_{n_1 n_2 n_3} = \frac{h^2}{8m} \left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2} \right)$$

- For a square box for $a=b=c=L$

$$E = \frac{h^2}{8m_e L^2} (n_1^2 + n_2^2 + n_3^2) = \frac{h^2}{8m_e L^2} n'^2$$

In 3D, we have 3 quantum numbers, each one arising from the boundary conditions along one of the coordinates.



Density of State

$$E = \frac{h^2}{8m_e L^2} (n_1^2 + n_2^2 + n_3^2) = \frac{h^2}{8m_e L^2} n'^2$$

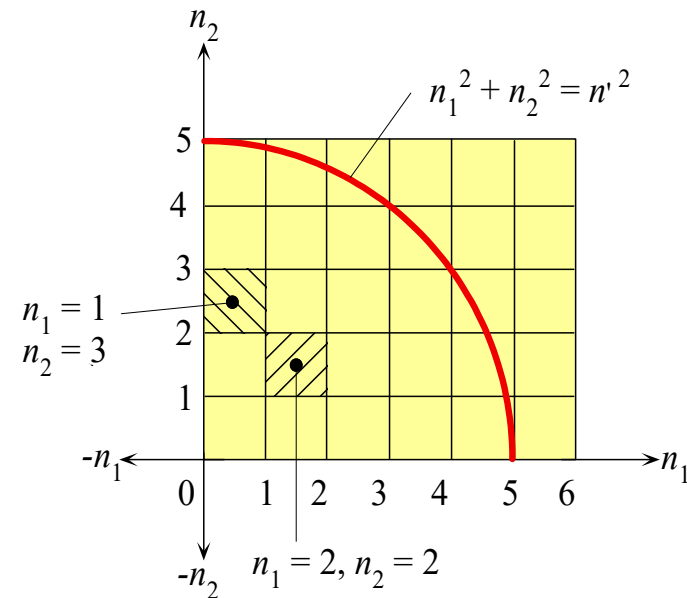
- The number of energy states depends on how many combination of n_1 , n_2 and n_3 satisfy:

$$n_1^2 + n_2^2 + n_3^2 \leq n'^2$$

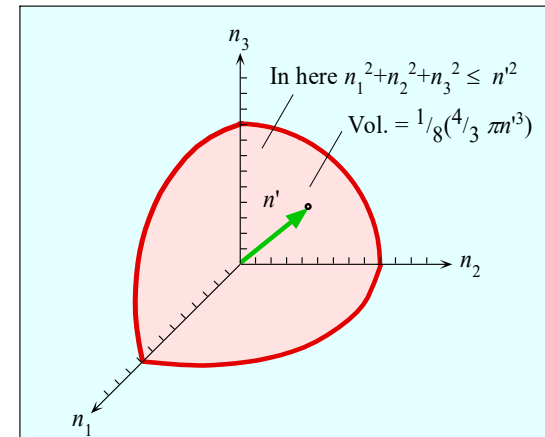
- In 2D space $S(n') = \frac{1}{4} \pi n'^2$

- In 3D space $S(n') = \frac{1}{8} \left(\frac{4}{3} \pi n'^3 \right) = \frac{1}{6} \pi n'^3$

- Take into accounts of spins $S(n') = \frac{1}{3} \pi n'^3$



Each state, electron wavefunction in the crystal, can be represented by a box at n_1, n_2 .



Density of State

➤ $n' = n_1^2 + n_2^2 + n_3^2 = \left(\frac{8m_e L^2 E}{h^2} \right)^{1/2}$ derive from $E = \frac{h^2}{8m_e L^2} (n_1^2 + n_2^2 + n_3^2) = \frac{h^2}{8m_e L^2} n'^2$

➤ Finally $S(E') = \frac{1}{3} \pi n'^3 = \frac{\pi L^3 (8m_e E')^{3/2}}{3h^3}$

➤ The number of states per unit volume is $S_v(E') = \frac{\pi (8m_e E')^{3/2}}{3h^3}$

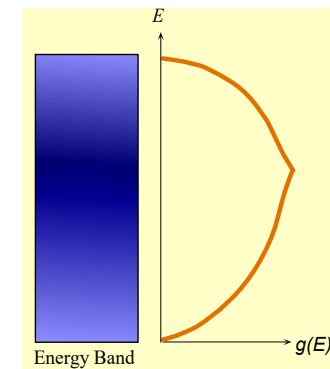
➤ Finally, the number of states in the energy interval E to $E+dE$ per unit volume is

$$g(E) = \frac{dS_v(E)}{dE} = 8\pi 2^{1/2} \left(\frac{m_e}{h^2} \right)^{3/2} E^{1/2}$$

➤ Approximate the true for band edges

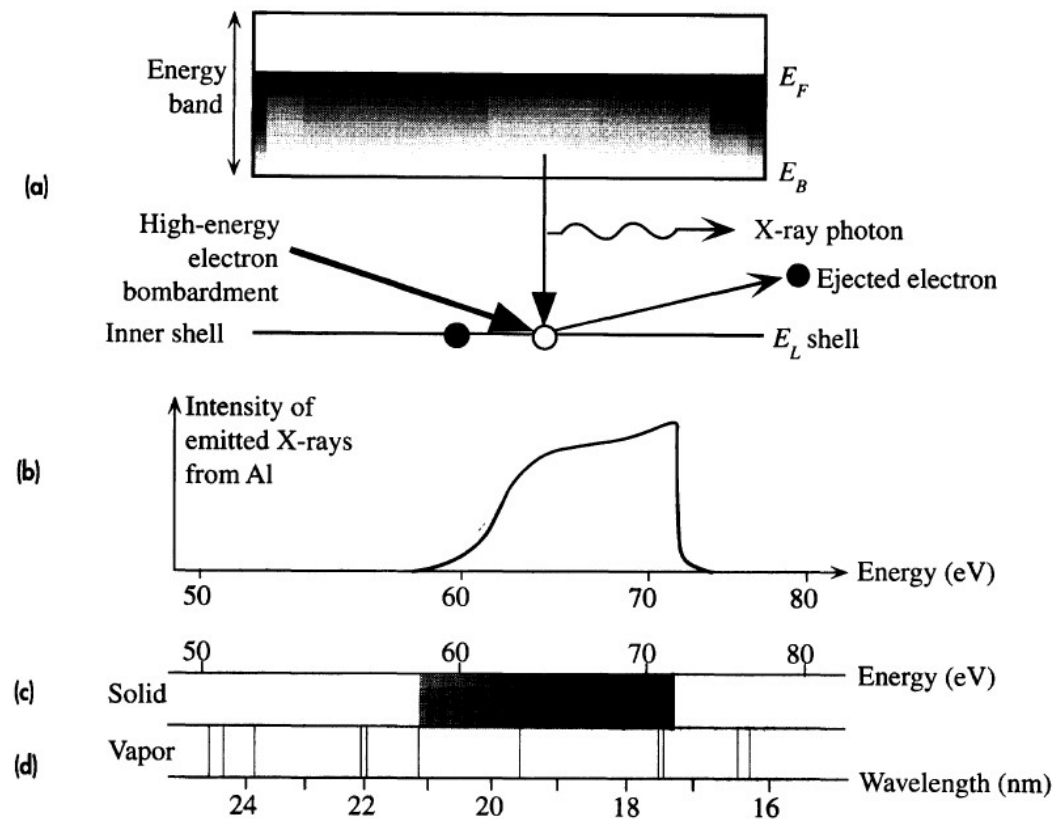
✓ Bottom of band $g(E) = 8\pi 2^{1/2} \left(\frac{m_e}{h^2} \right)^{3/2} (E - E_{bottom})^{1/2}$ increase with E

✓ Top of band $g(E) = 8\pi 2^{1/2} \left(\frac{m_e}{h^2} \right)^{3/2} (E_{top} - E)^{1/2}$ decrease with E



EXAMPLE 4.6

X-RAY EMISSION AND THE DENSITY OF STATES IN A METAL Consider what happens when a metal such as Al is bombarded with high-energy electrons. The inner atomic energy levels are not disturbed in the solid, so these inner levels remain as distinct single levels, each one localized to the parent atom. When an energetic electron hits an electron in one of the inner atomic energy levels, it knocks out this electron from the metal leaving behind a vacancy in the inner core as depicted in Figure 4.23a. An electron in the energy band of the solid can then fall down to occupy this empty state and emit a photon in the process. The energy difference between the energies in the band and the inner atomic level is in the X-ray range, so the emitted photon is an X-ray photon. Since electrons occupy the band from the bottom E_B to the Fermi level E_F , the



EXAMPLE 4.7

DENSITY OF STATES IN A BAND Given that the width of an energy band is typically ~ 10 eV, calculate the following, in per cm^3 and per eV units:

- The density of states at the center of the band.
- The number of states per unit volume within a small energy range kT about the center.
- The density of states at kT above the bottom of the band.
- The number of states per unit volume within a small energy range of kT to $2kT$ from the bottom of the band.

SOLUTION

- a The density of states, or the number of states per unit energy range per unit volume $g(E)$, is given by

$$g(E) = (8\pi 2^{1/2}) \left(\frac{m_e}{h^2} \right)^{3/2} E^{1/2}$$

which gives the number of states per cubic meter per Joule of energy. Substituting $E = 5$ eV, we have

$$g_{\text{center}} = (8\pi 2^{1/2}) \left[\frac{9.1 \times 10^{-31}}{(6.626 \times 10^{-34})^2} \right]^{3/2} (5 \times 1.6 \times 10^{-19})^{1/2} = 9.50 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1}$$

Converting to cm^{-3} and eV^{-1} , we get

$$\begin{aligned} g_{\text{center}} &= (9.50 \times 10^{46} \text{ m}^{-3} \text{ J}^{-1})(10^{-6} \text{ m}^3 \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ J eV}^{-1}) \\ &= 1.52 \times 10^{22} \text{ cm}^{-3} \text{ eV}^{-1} \end{aligned}$$

- b If δE is a small energy range (such as kT), then, by definition, $g(E) \delta E$ is the number of states per unit volume in δE . To find the number of states per unit volume within kT at the center of the band, we multiply g_{center} by kT or $(1.52 \times 10^{22} \text{ cm}^{-3} \text{ eV}^{-1})(0.026 \text{ eV})$ to get $3.9 \times 10^{20} \text{ cm}^{-3}$. This is not a small number!



c

At kT above the bottom of the band, at 300 K ($kT = 0.026$ eV), we have

$$g_{0.026} = (8\pi 2^{1/2}) \left[\frac{9.1 \times 10^{-31}}{(6.626 \times 10^{-34})^2} \right]^{3/2} (0.026 \times 1.6 \times 10^{-19})^{1/2}$$

$$= 6.84 \times 10^{45} \text{ m}^{-3} \text{ J}^{-1}$$

Converting to cm^{-3} and eV^{-1} we get

$$g_{0.026} = (6.84 \times 10^{45} \text{ m}^{-3} \text{ J}^{-1})(10^{-6} \text{ m}^3 \text{ cm}^{-3})(1.6 \times 10^{-19} \text{ J eV}^{-1})$$

$$= 1.10 \times 10^{21} \text{ cm}^{-3} \text{ eV}^{-1}$$

d

Within kT , the volume density of states is

$$(1.10 \times 10^{21} \text{ cm}^{-3} \text{ eV}^{-1})(0.026 \text{ eV}) = 2.8 \times 10^{19} \text{ cm}^{-3}$$

This is very close to the bottom of the band and is still very large.



TOTAL NUMBER OF STATES IN A BAND

- Based on the overlap of atomic orbitals to form the electron wavefunction in the crystal, how many states should there be in a band?
- Consider the density of states function

$$g(E) = (8\pi 2^{1/2}) \left(\frac{m_e}{h^2} \right)^{3/2} E^{1/2}$$

By integrating $g(E)$, estimate the total number of states in a band per unit volume, and compare this with the atomic concentration for silver. For silver, we have $E_{FO} = 5.5$ eV and $\Phi = 4.5$ eV. (Note that “state” means a distinct wavefunction, including spin.)

SOLUTION

- We know that when N atoms come together to form a solid, N atomic orbitals can overlap N different ways to produce N orbitals or $2N$ states in the crystal, since each orbital has two states, spin up and spin down. These states form the band.
- For silver, $E_{FO} = 5.5$ eV and $\Phi = 4.5$ eV, so the width of the energy band is 10 eV. To estimate the total volume density of states, we assume that the density of states $g(E)$ reaches its maximum at the center of the band $E = E_{\text{center}} = 5$ eV. Integrating $g(E)$ from the bottom of the band, $E = 0$, to the center, $E = E_{\text{center}}$, yields the number of states per unit volume up to the center of the band. This is half the total number of states in the whole band, that is, $\frac{1}{2}S_{\text{band}}$, where S_{band} is the number of states per unit volume in the band and is determined by

$$\frac{1}{2}S_{\text{band}} = \int_0^{E_{\text{center}}} g(E) dE = \frac{16\pi 2^{1/2}}{3} \left(\frac{m_e}{h^2} \right)^{3/2} E_{\text{center}}^{3/2}$$



or

$$\begin{aligned}\frac{1}{2}S_{\text{band}} &= \frac{16\pi 2^{1/2}}{3} \left[\frac{9.1 \times 10^{-31} \text{ kg}}{(6.626 \times 10^{-34} \text{ J s})^2} \right]^{3/2} (5 \text{ eV} \times 1.6 \times 10^{-19} \text{ J/eV})^{3/2} \\ &= 5.08 \times 10^{28} \text{ m}^{-3} = 5.08 \times 10^{22} \text{ cm}^{-3}\end{aligned}$$

Thus

$$S_{\text{band}} = 10.16 \times 10^{22} \text{ states cm}^{-3}$$

We must now calculate the number of atoms per unit volume in silver. Given the density $d = 10.5 \text{ g cm}^{-3}$ and the atomic mass $M_{\text{at}} = 107.9 \text{ g mol}^{-1}$ of silver, the atomic concentration is

$$n_{\text{Ag}} = \frac{d N_A}{M_{\text{at}}} = 5.85 \times 10^{22} \text{ atoms cm}^{-3}$$

As expected, the density of states is almost twice the atomic concentration, even though we used a crude approximation to estimate the density of states.



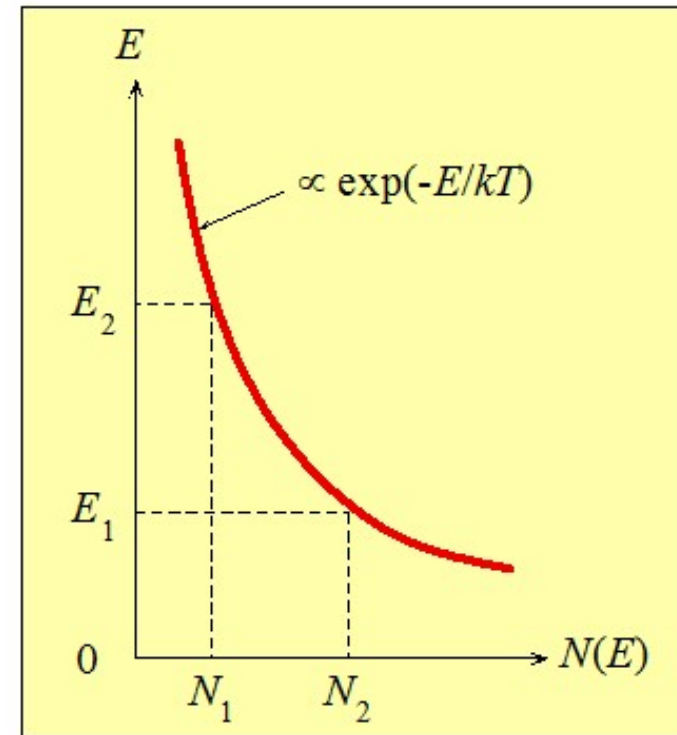
Boltzmann Classical Statistics

- If $f(E)$ is the probability that a state with energy E is occupied, then the number of electron is $n = \int_{band} f(E) g(E) dE$
- How to find $f(E)$?

Boltzman function

$$P(E) = A \exp\left(-\frac{E}{kT}\right)$$

- ❑ Number of electrons decreases exponentially with energy
- ❑ At the bottom of the band, E small $\rightarrow$ more electrons

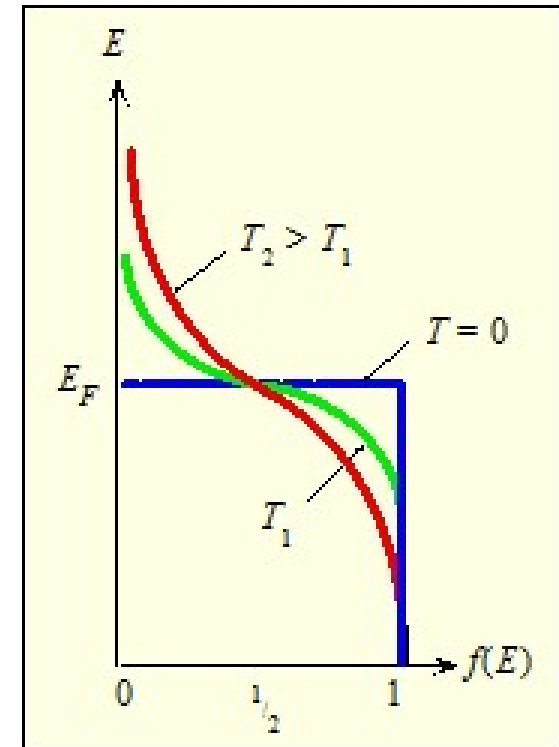


Fermi-Dirac Statistics

Fermi-Dirac function

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

- Fermi level: energy level at which the probability of finding an electron, for $T > 0$ K, is exactly 1/2
- At 0 K every available energy state up to E_F is filled with electrons, all states above E_F are empty
- At $E - E_F \gg kT$, Fermi-Dirac function is equal to Boltzmann function $f(E) = \exp\left(\frac{E - E_F}{kT}\right)$
- Hole probability function: $1 - f(E)$

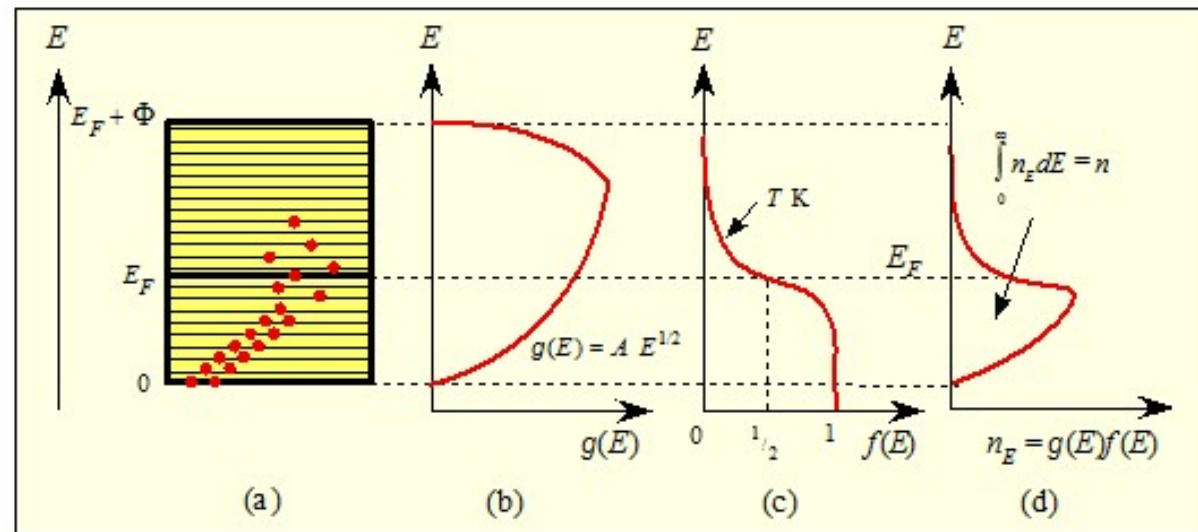


Number of Electrons in Metals

- So far we know $g(E)$ and $f(E)$, how to obtain $n(E)$?

$$g(E) = \frac{dS_v(E)}{dE} = \left(8\pi 2^{1/2}\right) \left(\frac{m_e}{h^2}\right)^{3/2} E^{1/2} \quad f(E) = \frac{1}{1 + A \exp\left(\frac{E - E_F}{kT}\right)}$$

- $n(E)$: number of electrons per unit energy per unit volume, is simply the product of $g(E)*f(E)$



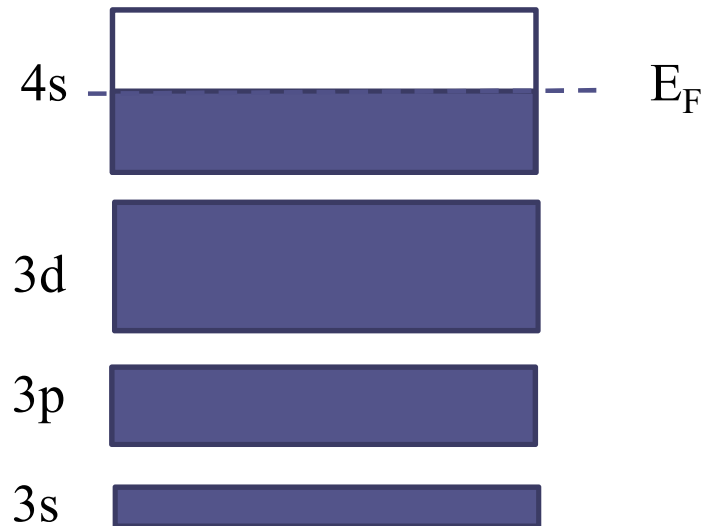
Number of electrons per unit volume is
$$n = \int_0^{\text{top of band}} n(E) dE = \int_0^{\text{top of band}} f(E) g(E) dE$$



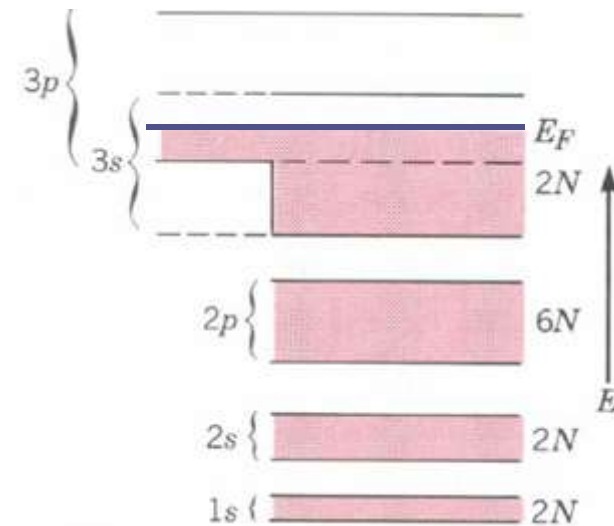
Conductivity and Fermi Level in Metals

- Only those electrons in a small range ΔE around the Fermi level contribute to conduction, why?
- $\sigma = nq\mu = \frac{e^2 n \tau}{m_e}$, the larger carrier concentration, the higher conductivity
- $n_F = g(E_F) \Delta E = g(E_F) e v_F \tau E$, the concentration of electrons mainly depending on density of state near Fermi level.
- Cu and Mg, whose conductivity is larger?

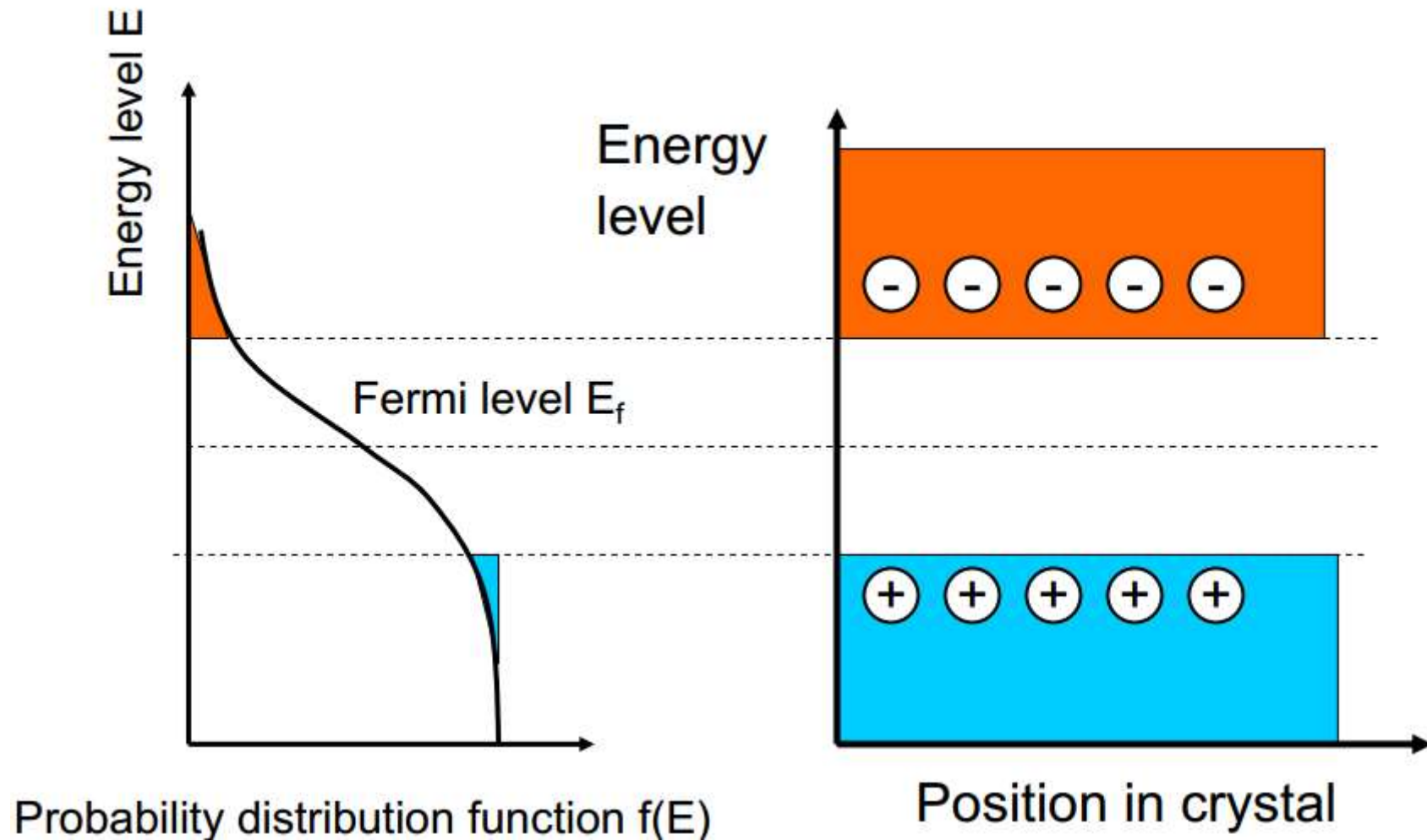
Cu: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$



Mg: $1s^2 2s^2 2p^6 3s^2$



Fermi Level for Intrinsic Semiconductor

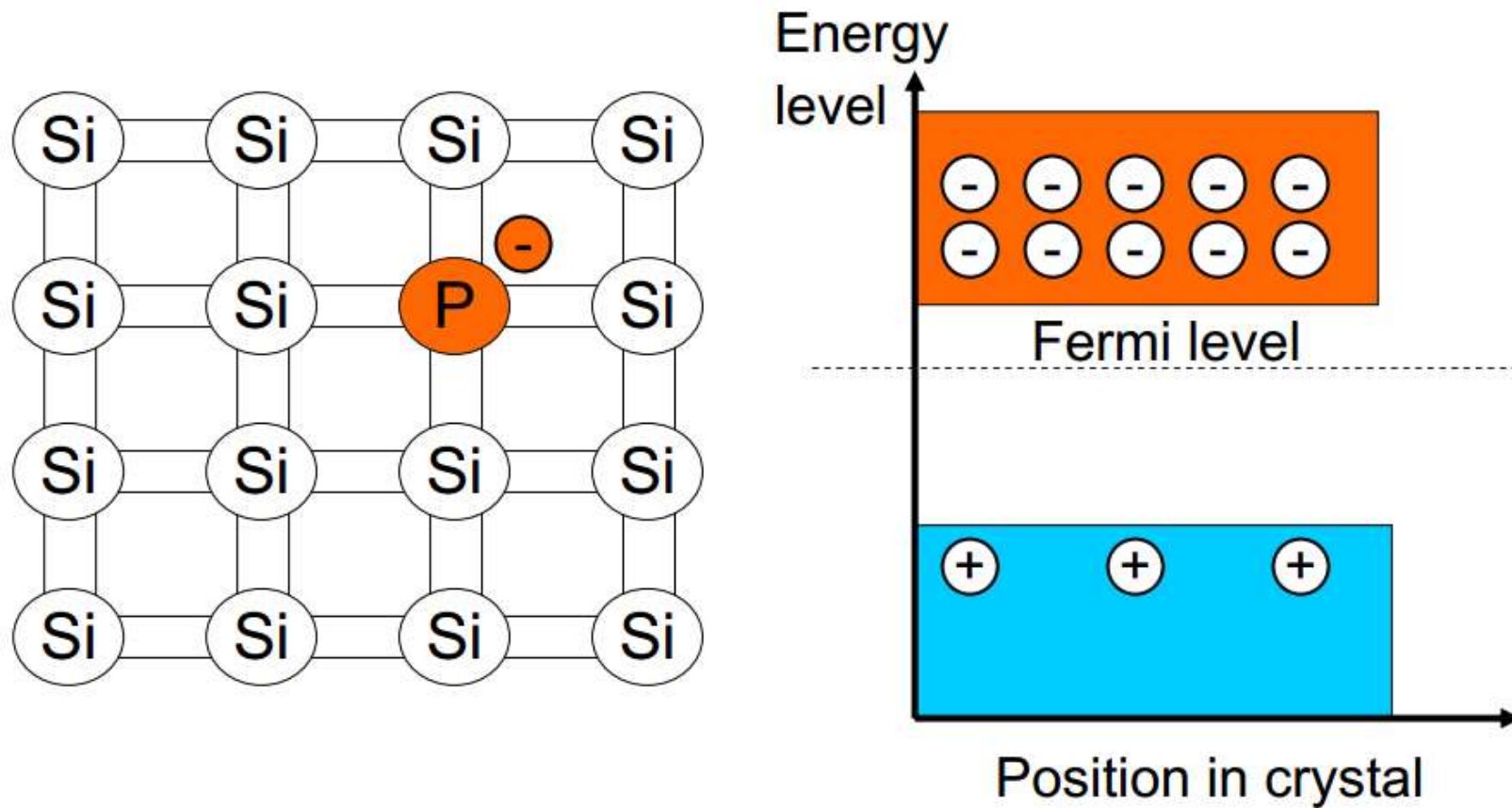


$$n = \int_{band} f(E) g(E) dE$$



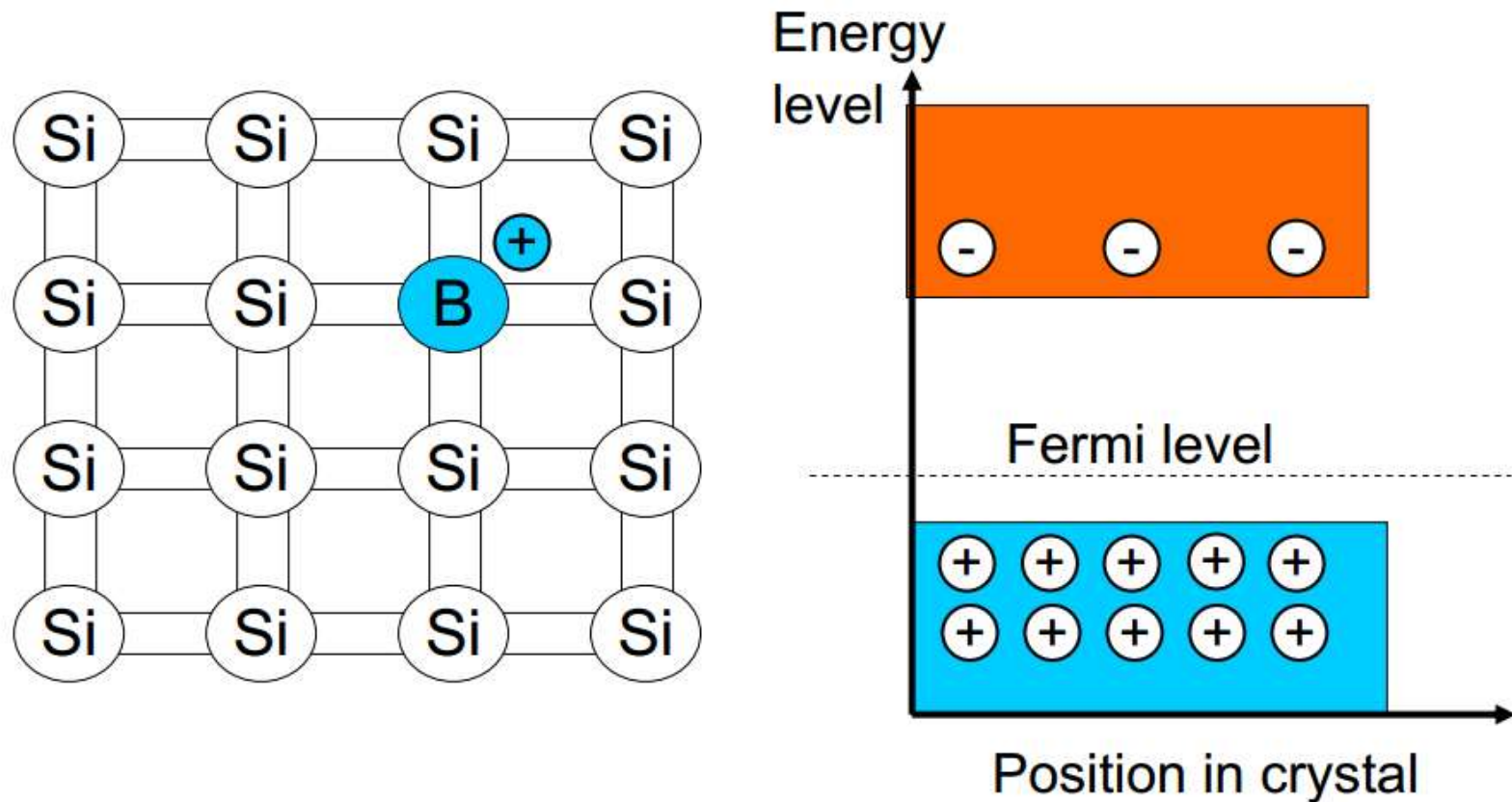
Fermi Level for *n*-Type Semiconductor

Doping: n-type

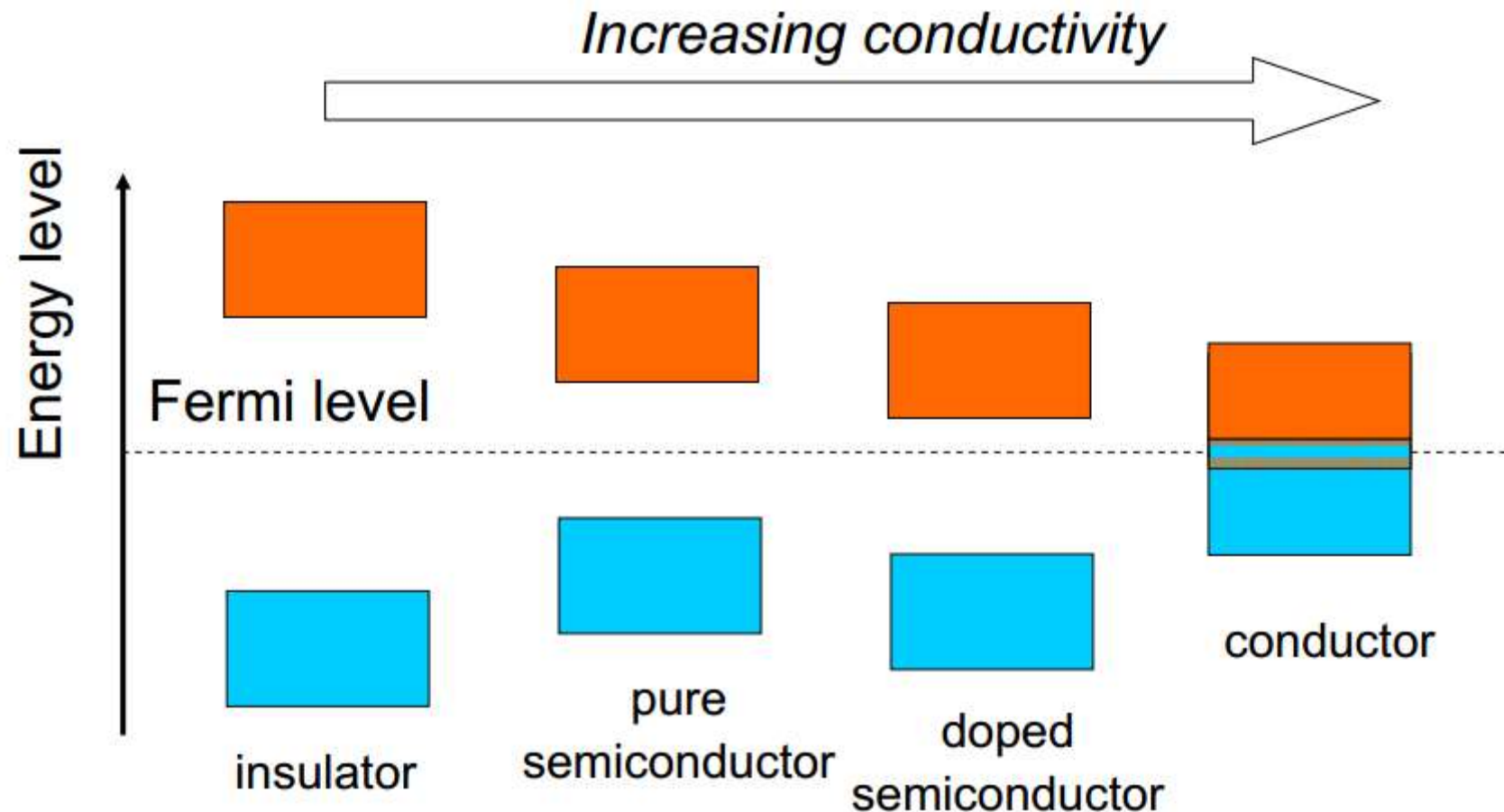


Fermi Level for *p*-Type Semiconductor

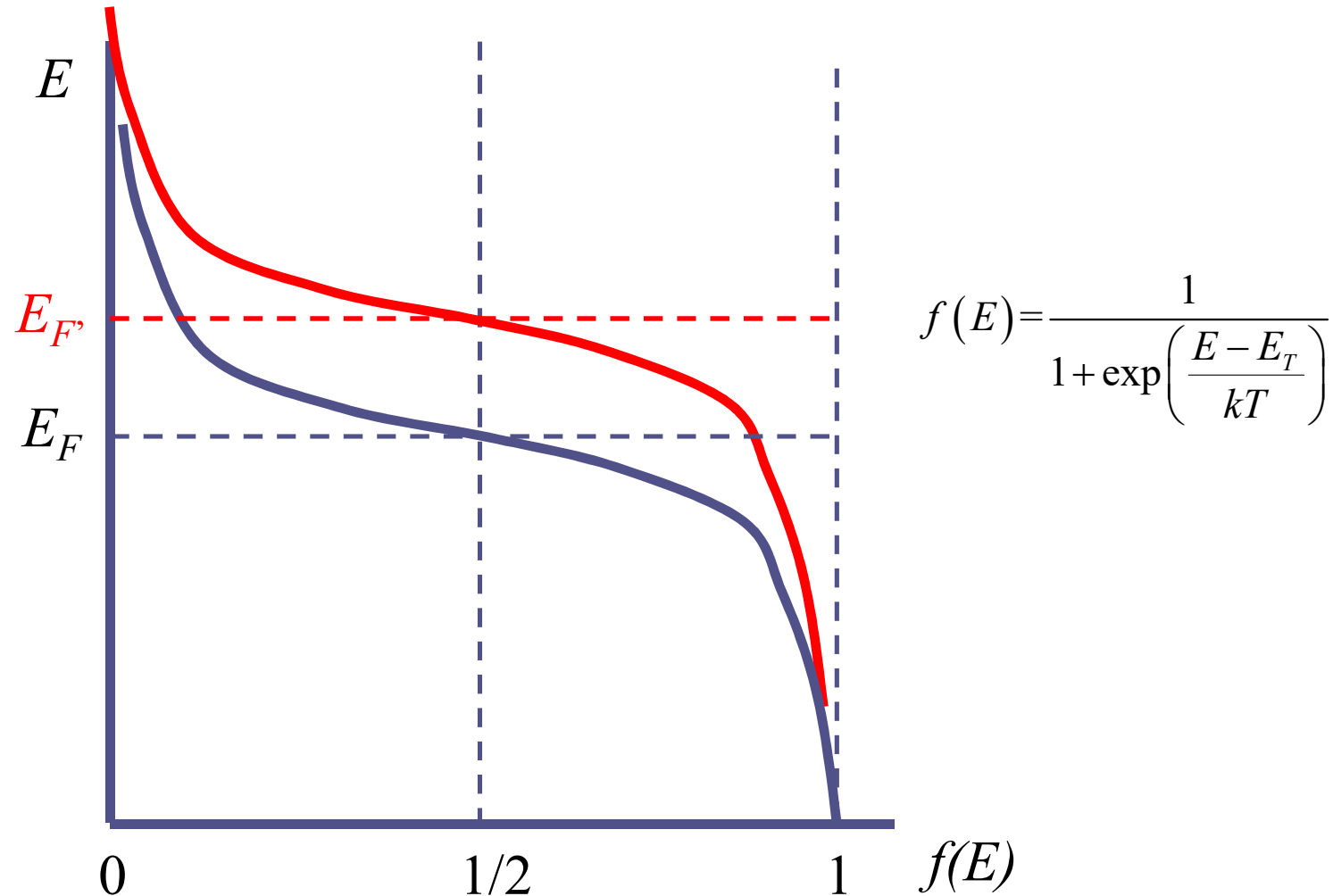
Doping: p-type



Insulator, Semiconductor and Conductor



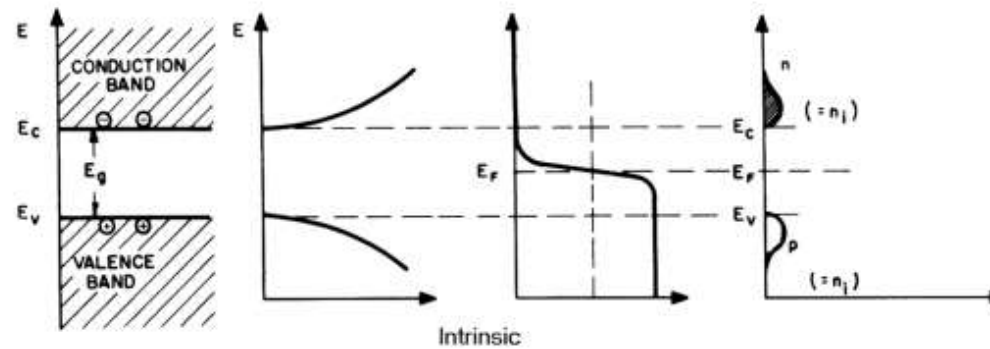
How to Draw the Fermi Distribution Function



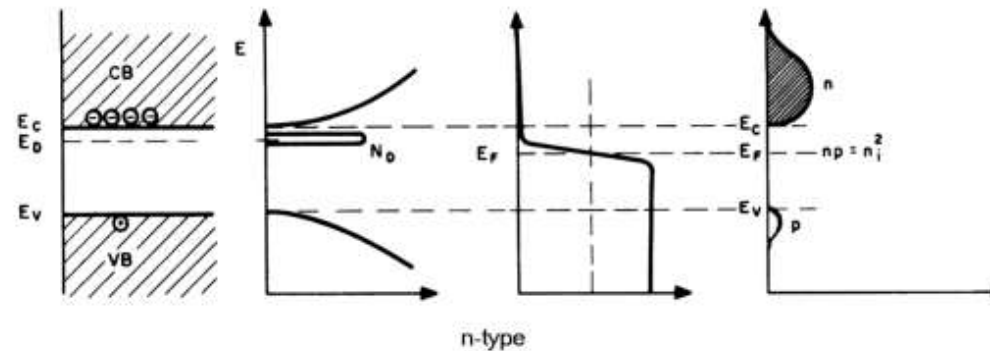
Carriers in Semiconductor

Additional Dopant States

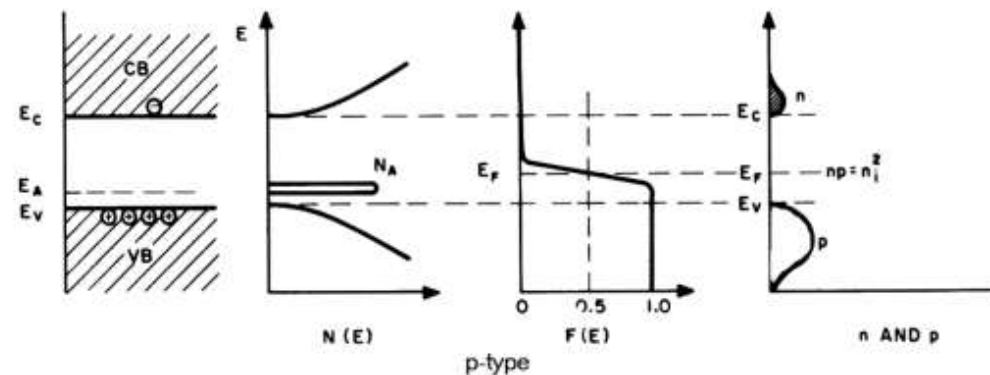
Intrinsic:
Equal number
of electrons
and holes



n-type: more
electrons than
holes



p-type: more
holes than
electrons



Summary

- E - k diagram of solids.
- Origin of band gap: standing wave in the bottom and the top of the energy band.
- Effective mass and velocity from E - k diagram.
- Density of states.
- Fermi-Dirac distribution for semiconductors.
- Fermi level for semiconductors.
- Carrier concentration

